Total No. of Printed Pages: 02

SUBJECT CODE NO: - RNP-8011

FACULTY OF SCIENCE AND TECHNOLOGY

S.E. (Mechanical) (NEP) Sem - III

Examination November / December 2025 (To be held in February-2026)

Machine Drawing

[Time: 2:00 Hours] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1. **Part A: Question No 1 is compulsory.**
2. **Attempt any four questions from remaining Part B: Q.2 to Q.7.**

Part A

Q1 Solve any 5 following questions: **10**

a) The development of a prism consists of:
A) A series of rectangles B) A circle
C) A triangle D) A parabola

b) The curve of interpenetration between two cylinders is generally:
A) Straight line B) Ellipse C) Circle D) Spiral

c) An auxiliary view helps to obtain:
A) Hidden view B) True shape and size
C) Sectional view D) Simplified drawing

d) A helix is a curve generated by:
A) A point moving around and along a cylinder simultaneously
B) A rolling circle on a straight line
C) A circle on a cone
D) A parabola

e) Surface finish symbols are given to indicate:
A) Material type B) Type of machining process
C) Sectional area D) Thread direction

f) A rivet joint is a type of:
A) Temporary joint B) Permanent joint
C) Sliding joint D) Flexible joint

g) The key in a shaft is used to:
A) Prevent relative motion between shaft and hub
B) Increase torque

C) Reduce friction
D) Support load

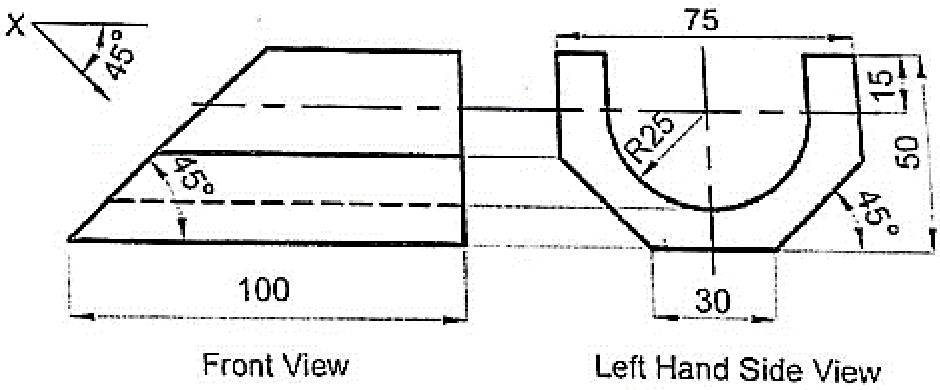
Part B

Solve any four from the following (5 Marks Each)

Q2 A cylinder of base diameter 50mm and axis 70 mm is resting on ground with its axis vertical. It is cut by a section plane perpendicular to the VP inclined at 45 degree to the HP passing through the top of generator and cuts the other entire generator. Draw the development of its lateral surface. **5**

Q3 A vertical cylinder of 80mm dia is completely penetrated by another cylinder of 60mm dia, their axes bisecting each other at right angles. Draw their projection showing curves of penetration assuming the axis of the penetrating cylinder to be parallel to VP (assume suitable length of the prism.) **5**

Q4 Fig shows a front view and left hand side view of an object. Draw the given views and project an auxiliary top view looking in the direction of X. **5**



Q5 Draw an involute of a circle of 50mm diameter. Also draw tangent and normal on the curve. **5**

Q6 Draw and label the parts of a cotter joint. **5**

Q7 Sketch and label different types of bearings with their applications. **5**